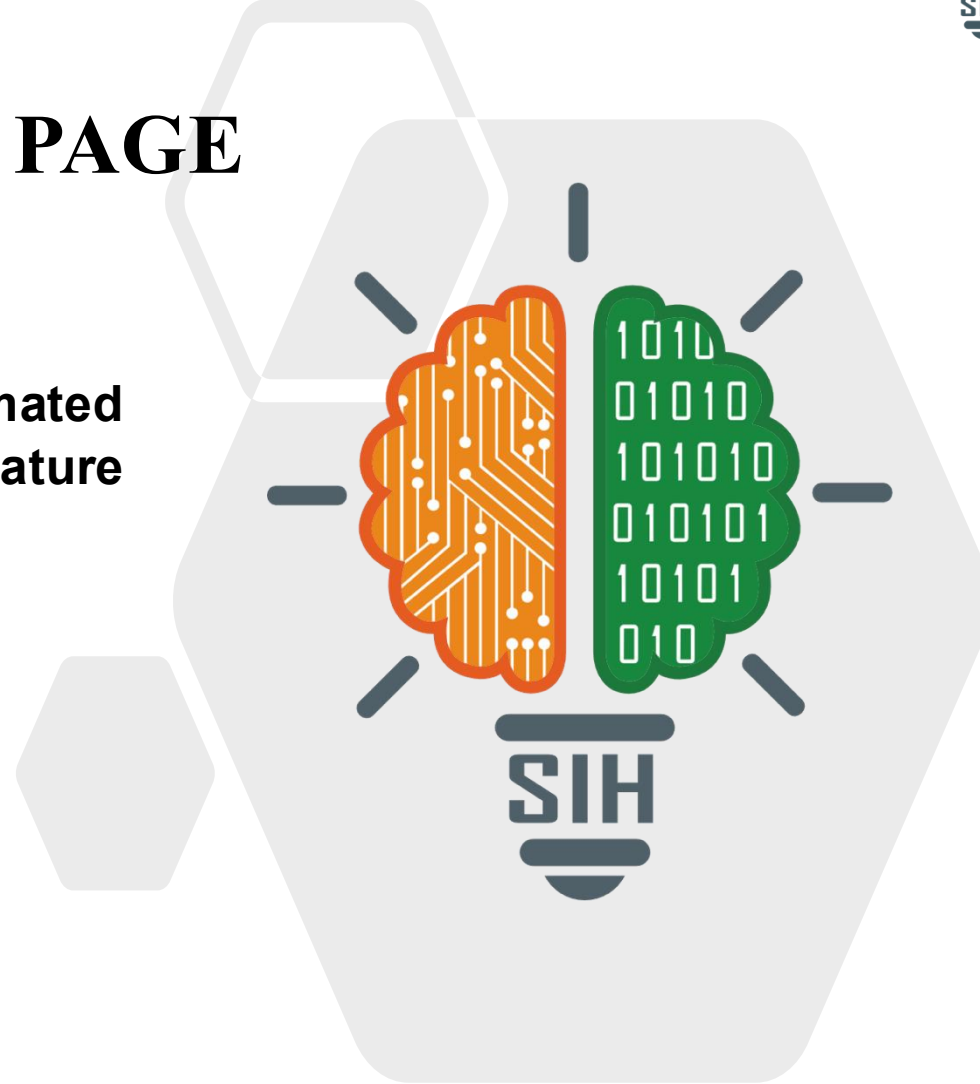


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26012
- Problem Statement Title- AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery
- Theme- Smart Automation
- PS Category- Software
- Team ID- 126010
- Team Name- SUPER6



GeoParcel AI — drone & satellite imagery → verification-ready building-parcel maps

> Proposed Solution

An end-to-end AI pipeline that turns an orthomosaic (or a satellite map) into clean, georeferenced building-footprint polygons — the parcel candidates a surveyor verifies instead of tracing by hand.

- **Scale-aware AI segmentation:** U-Net with ResNet-34 encoder labels every pixel as building / background; overlapping 512-px tiled inference handles images of any size.
- **Works from drone to satellite:** multi-scale model trained at 7–45 cm/pixel; GeoTIFF resolution is read automatically.
- **Rectilinear regularisation:** jagged masks become right-angled polygons ($\sim 27 \times$ fewer vertices) through orientation snapping, least-squares edge refits and an IoU fidelity guard.
- **GIS-ready output:** GeoJSON in the image's own CRS (UTM / WGS84), ready for QGIS, Web-GIS and the NAKSHA workflow.
- **Surveyor in the loop:** every polygon is tagged “candidate – needs field verification”; corrections feed retraining.

> How it addresses the problem

- **Removes the bottleneck:** manual digitisation sits between the drone flight and the property record; we automate it.
- **Consistent & auditable:** the same model and rules for every ward, with per-polygon review status.
- **Scales nationally:** NAKSHA covers 152 ULBs in 26 States + 3 UTs; SVAMITVA has surveyed 3.29 lakh villages.

> Innovation & uniqueness

- **Multi-resolution:** one model for 7 cm drone and ~ 45 cm satellite imagery (IoU 0.14 $\rightarrow$ 0.61 at 45 cm).
- **Shape-aware post-processing:** polygons a surveyor can edit, not pixel blobs.
- **Proven, not proposed:** working prototype, open code, public live demo.
- **Low cost:** trained on one laptop GPU; runs on free cloud hardware.



Raw AI contour

~ 240 vertices per building

GeoParcel AI polygon

~ 10 vertices, right-angled

0.80

pixel IoU on an unseen neighbourhood

27 $\times$

fewer vertices per polygon

0.14 $\rightarrow$ 0.61

IoU on 45 cm satellite-like imagery

Working prototype — try it live

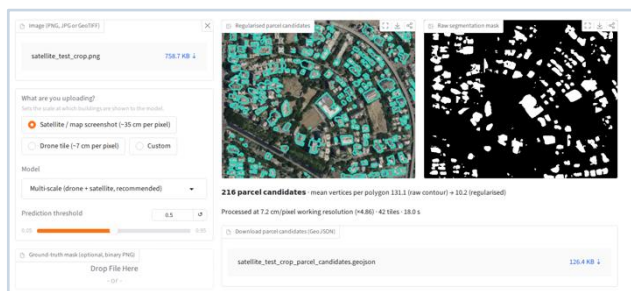
Demo: huggingface.co/spaces/omeow/geoparcels-ai

Code: github.com/OMMEOW/SlH26_PS26012

Upload a drone tile or satellite screenshot $\rightarrow$ polygons + GeoJSON download.

> Technologies to be used

Layer	Technology
AI / ML	PyTorch · segmentation_models_pytorch (U-Net, ResNet-34) · timm · Dice + BCE loss
Geospatial	Rasterio / GDAL · GeoPandas · Shapely · PyProj (WGS84 ↔ UTM)
Vision & maths	OpenCV · NumPy · SciPy · Albumentations (augmentation)
Applications	Gradio (public web app) · Streamlit (operator console)
Deployment	Hugging Face Spaces (ZeroGPU) · Docker · GitHub Releases (weights)
Output standard	GeoJSON in source CRS → QGIS / Web-GIS / NAKSHA portal
Data	Open Cities AI Challenge drone orthomosaics, 4–7 cm/pixel (ODbL)



Live web app: upload → polygons → GeoJSON

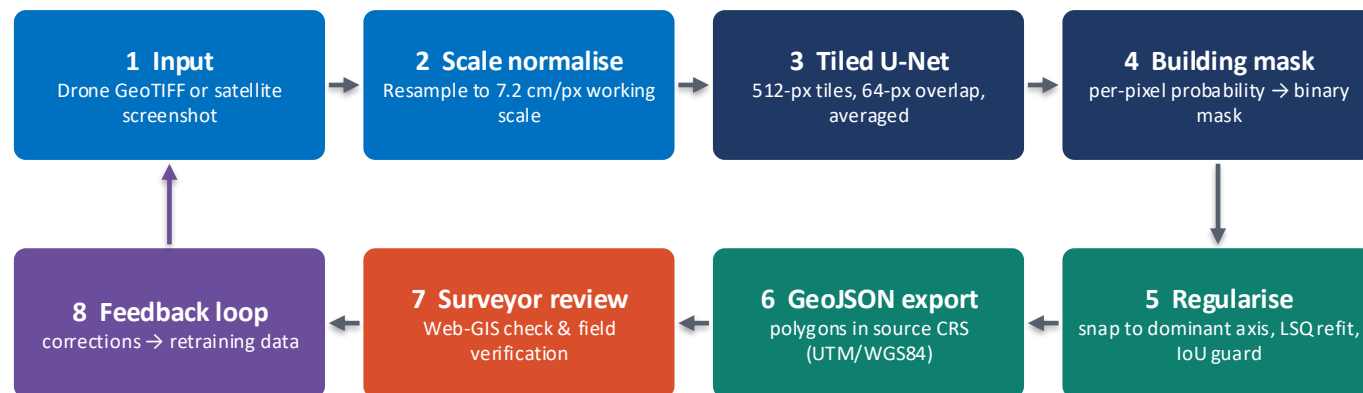
Workflow fit

Drone survey → orthomosaic
→ GeoParcel AI → surveyor
→ land record

Speed

640 × 520 px satellite
screenshot (42 tiles): ~4 s on
a shared GPU, ~18 s on CPU.

> Methodology & process flow



> Model development

CRS-aligned tiling

WGS84 labels reprojected to
UTM before rasterising (else
masks are empty)

Base training

5,008 tiles · 7.2 cm · 10
epochs · best val IoU 0.63

Multi-scale fine-tune

2 more scenes at 7.2–45 cm ·
JPEG/blur/colour aug · 6
epochs

Evaluation

held-out tiles · unseen 396 m
scene · accuracy per
resolution

Project links (working prototype)

- Live demo: <https://huggingface.co/spaces/omeow/geoparcel-ai>
- GitHub: https://github.com/OMMEOW/SIH26_PS26012
- Model weights: https://github.com/OMMEOW/SIH26_PS26012/releases/tag/weights-v1



FEASIBILITY AND VIABILITY



> Feasibility (already demonstrated)

- **Technical:** full pipeline built and tested on real drone imagery. Pixel IoU 0.60 on 502 held-out tiles; 0.80 on a never-seen 396 m × 396 m neighbourhood; 427 GeoJSON candidates produced end to end.
- **Economic:** open-source stack, no licence fees; trained on a single laptop GPU; hosted on free cloud tier.
- **Operational:** standard inputs (GeoTIFF) and outputs (GeoJSON) — slots into existing GIS tools; simple web UI for field teams.
- **Deployable anywhere:** Docker image for State data centres / NIC cloud, so imagery need not leave government servers.

> Viability

- **Clear demand:** NAKSHA urban survey (152 ULBs, 26 States + 3 UTs) and SVAMITVA (3.29 lakh villages surveyed) both generate drone orthomosaics that need digitising.
- **Incremental adoption:** starts as decision support for surveyors; accuracy improves with every verified correction.

SUPPORTING FACTS (measured on our prototype)

- Regularisation: 213 → 7.9 vertices per polygon; right-angle corners 17% → 35%; polygon IoU preserved (0.47 → 0.46).
- Multi-scale model: pixel IoU 0.61 → 0.67 at 7.2 cm and 0.14 → 0.61 at 45 cm on an unseen scene.
- No loss on the original drone validation set (0.642 → 0.643).

> Challenges, risks & mitigation

Challenge	Risk	Mitigation strategy
Indian building stock (roofs, high-rise towers)	Missed buildings	Fine-tune on NAKSHA / SVAMITVA orthomosaics; active learning from surveyor edits
Touching buildings with shared walls	Merged parcels	Boundary-aware loss / instance segmentation; split tool in review UI
Mixed image resolutions & sources	Accuracy drop	Multi-scale training (done); resolution read from GeoTIFF metadata
Legal acceptance of AI output	Disputes	Outputs are candidates only; mandatory field verification & audit trail
Data security & sovereignty	Data leakage	On-premise Docker deployment; no third-party APIs needed
Scale (crores of parcels)	Compute cost	Tiled batch inference on GPU nodes; CPU fallback; process per ward

Roll-out plan

- **Pilot:** one NAKSHA ULB — compare against manual digitisation (time, accuracy).
- **Adapt:** fine-tune on Indian orthomosaics + surveyor corrections.
- **Scale:** State-level deployment, then plot boundaries (walls, fences, roads).

> Potential impact on the target audience

- **DoLR / State Revenue & Survey depts:** faster conversion of drone surveys into digital land records.
- **Urban Local Bodies:** up-to-date building footprints for property tax, planning and encroachment checks.
- **Surveyors:** verify and correct instead of tracing every outline; more wards covered per team.
- **Citizens:** quicker property cards → easier loans, fewer boundary disputes.
- **Banks & planners:** standard GeoJSON data for credit, infrastructure and disaster planning.
- **Disaster response:** rapid building inventories from post-event drone flights (the Open Cities data was built for this).

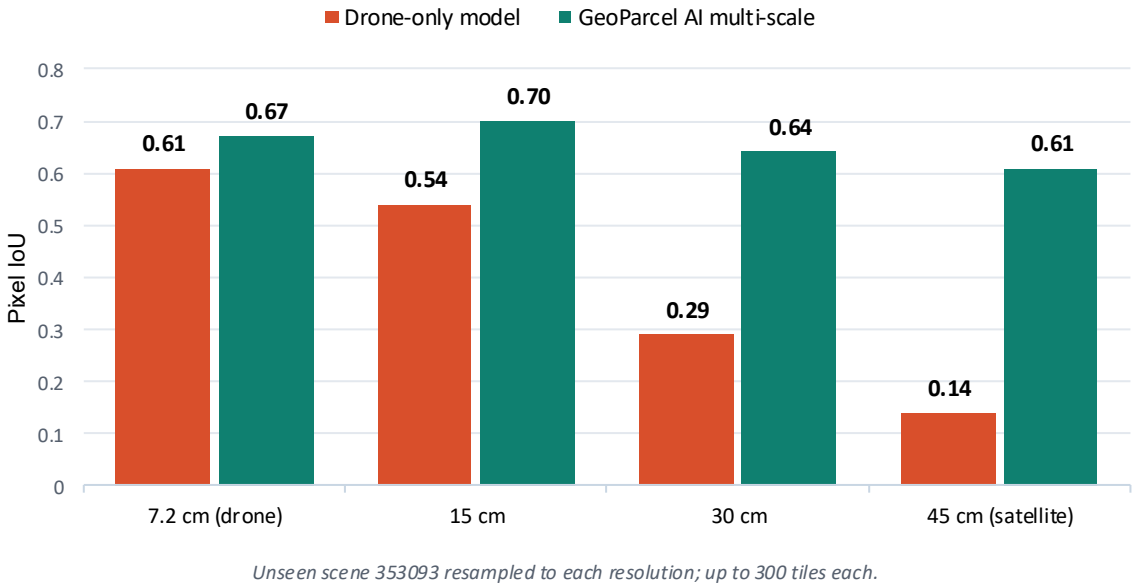


Indian city, satellite map (~35 cm/px)
Left: drone-only model — most buildings missed.
Right: multi-scale model — mid-rise and large flat roofs found.
No ground truth here: qualitative.

> Benefits of the solution

Type	Benefit	How GeoParcel AI delivers it
Social	Faster, more transparent property records; fewer boundary disputes	Uniform, auditable polygons with surveyor sign-off feed SVAMITVA / NAKSHA property cards
Economic	Surveyor time moves from tracing to verification; better property-tax base for ULBs	Automatic footprints + GeoJSON export; open-source, runs on commodity hardware
Environmental	Fewer repeat field visits; reuses imagery already captured	Works on existing drone and satellite orthomosaics — no new flights needed
Governance	Standard, machine-readable land data	Georeferenced GeoJSON in source CRS, ready for Web-GIS and Digital India Land Records

> Measured accuracy: drone → satellite resolution





RESEARCH AND REFERENCES



> Research papers & methods

a. Ronneberger, O., Fischer, P., Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI. arxiv.org/abs/1505.04597

b. He, K., Zhang, X., Ren, S., Sun, J. (2016). Deep Residual Learning for Image Recognition. CVPR. arxiv.org/abs/1512.03385

c. Douglas, D., Peucker, T. (1973). Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. The Canadian Cartographer, 10(2).

d. Zero-Shot Building Footprint Extraction and Regularization Based on Segment Anything Model with Mesh Model.

e. UrbanSAM: Learning Invariance-Inspired Adapters for Segment Anything Models in Urban Construction.

f. The Segment Anything Model (SAM) for Remote Sensing Applications: From Zero to One Shot.

g. Enhancing Conventional Land Surveying for Cadastral Documentation in Romania with UAV Photogrammetry and SLAM.

> Data, programmes & tools

- Open Cities AI Challenge dataset — GFDRR Labs (2020), drone imagery + building labels, ODbL-1.0. source.coop/open-cities
- NAKSHA urban land survey pilot — 152 ULBs, 26 States + 3 UTs (DoLR, Feb 2025). newsonair.gov.in
- SVAMITVA — drone survey done in 3.29 lakh of 3.44 lakh villages (PIB, 11 Mar 2026). [PIB India](https://pibindia.gov.in)
- segmentation_models_pytorch, Rasterio, GeoPandas, Shapely, Gradio (open source).

Project links (live prototype)

- Live demo: <https://huggingface.co/spaces/omeow/geoparcel-ai>
- GitHub: https://github.com/OMMEOW/SIH26_PS26012

> How we compare

Capability	GeoParcel AI	Manual GIS tracing	Raw AI mask
Automatic building extraction	✓	✗	✓
Clean right-angled polygons	✓	✓	✗
Georeferenced GeoJSON output	✓	✓	✗
Handles drone + satellite resolutions	✓	✓	~
Same rules for every ward / operator	✓	~	✓
Surveyor verification step	✓	✓	✗

“Raw AI mask” = a segmentation model without our regularisation and GIS export.

> Research flow

